

# NETWORKS AND COMPLEXITY

## Solution 16-1

*This is an example solution from the forthcoming book *Networks and Complexity*.*

*Find more exercises at <https://github.com/NC-Book/NCB>*

### Ex 16.1: Geometric dynamics [Lvl. 1]

Starting from the initial condition  $\delta(0) = \delta_0$ , integrate

$$\dot{\delta} = a\delta^2.$$

#### Solution

We separate the variables, by writing the differential equation as

$$\delta^{-2}d\delta = a dt. \tag{1}$$

Then we integrate

$$\int \delta^{-2}d\delta = \int a dt. \tag{2}$$

$$-\delta^{-1} = at + C \tag{3}$$

$$\delta^{-1} = -at - C \tag{4}$$

$$\tag{5}$$

which we can solve for

$$\delta(t) = \frac{1}{-at - C}. \tag{6}$$

To fix the constant of integration, we recall the initial condition

$$\delta_0 = \frac{1}{-C} \tag{7}$$

which implies

$$-C = \delta_0^{-1} \tag{8}$$

Hence we can write the solution to the initial value problem as

$$\delta(t) = \frac{1}{\delta_0^{-1} - at} \tag{9}$$